Applying Continuum Mechanics To A Solid Wire

(Dated: May 2020)

BACKGROUND

Continuum Mechanics

Chapter 16 of Classical Mechanics by John Taylor offers an introduction into continuum mechanics. The book previously has looked at the mechanics of point masses, and the mechanics of rigid bodies. Evaluating rigid bodies, we assumed that for a single body, the mass would be distributed in some way such that the relative positions of the mass would not change. In other words, a single body would maintain the same shape and density distribution while being acted on by forces.

In continuum mechanics there are an infinite number of parts and coordinates corresponding to infinitesimal pieces of mass. Continuum mechanics looks at cases where the mass of a body distributed over some region can change in some continuous way. In other words, individual pieces of the body will be affected depending on the time and their location, and can change relative to each other. We could think of a wave travelling through a string, or a bending metal beam.

Surface Forces

To evaluate a three dimensional continuous system, you start with a (likely infinitesimal) volume element dV, which can be chosen as some arbitrary shape. The volume dV is bounded by a closed surface, and we use a unit vector n (normal to the surface and pointing outward) to orient the surface.

Two types of forces can act on dV. Forces that are proportional to dV, such as gravity, are labelled volume forces. We are familiar with a lot of these and do not need to focus on them. Surface forces are proportional to area elements dA. The three surface forces to focus on are pressure, tension, and shear. Tension is a force pointing the same direction as n, pressure points opposite that, and shear will instead point parallel the surface.

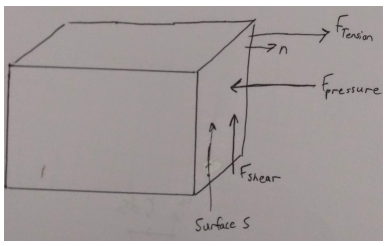

FIG. 1: Forces

Stress

Stress is the ratio of the surface force F to the area it is acting upon. In the specific case of a wire, stress is defined as

$$\text{stress} = \text{tension/area},$$

with area being defined as the cross-section of the wire.

To express the surface forces acting on the body, we use the stress tensor. Consider the surface force $\mathbf{F}$ acting on $d\mathbf{A}$ (composed of a scalar surface element dA oriented by $\mathbf{n}$). It can be shown that $\mathbf{F}(d\mathbf{A})$ is a linear function of $d\mathbf{A}$, which can be be written in the matrix form

$$F(d\mathbf{A}) = \mathbf{\Sigma} d\mathbf{A}. \tag{1}$$

$\mathbf{\Sigma}$ is the **stress tensor** (not a summation sign). It can be written as the matrix

$$\mathbf{\Sigma} = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{bmatrix} \tag{2}$$

One important property about the stress tensor is that it is always symmetric, that is, $\sigma_{ij} = \sigma_{ji}$.

Strain

We define strain as the resulting fractional change in dimensions (or deformation) of a body undergoing stress. For a wire,

$$\text{strain} = dl/l,$$

Where dl/l is the fractional change in length.

To describe the displacements undertaken in the body, we use the strain tensor for a solid. First, we picture each small mass dV being displaced from an original position r by u(r). Since we are trying to measure deformation, we are looking to find the change in separation between points

$$d\mathbf{u} = \mathbf{D} d\mathbf{r}$$

Where $\mathbf{D}$ is the derivatives tensor that measures how quickly $\mathbf{u}(r)$ changes in the body.

$$\mathbf{D} = \begin{bmatrix} \partial u_{11}/\partial r_{11} & \partial u_{12}/\partial r_{12} & \partial u_{13}/\partial r_{13} \\ \partial u_{21}/\partial r_{21} & \partial u_{22}/\partial r_{22} & \partial u_{23}/\partial r_{23} \\ \partial u_{31}/\partial r_{11} & \partial u_{32}/\partial r_{12} & \partial u_{33}/\partial r_{33} \end{bmatrix} \tag{3}$$

To measure deformation, we must also be careful that the whole body does not move in a rigid rotation. Rigid rotation is undergone when $\mathbf{D}$ is an antisymmetric matrix, which can be written in the general form

$$\mathbf{D} = \begin{bmatrix} 0 & a_{12} & a_{13} \\ -a_{21} & 0 & a_{23} \\ -a_{31} & -a_{32} & 0 \end{bmatrix} \tag{4}$$

We can use a property of matrices to split $\mathbf{D}$ into antisymmetric ($\tilde{\mathbf{D}}$) and symmetric $\mathbf{E}$ parts.

$$\mathbf{E} = (1/2)(\mathbf{D} + \tilde{\mathbf{D}}) \tag{5}$$

$\mathbf{E}$ is the **strain tensor**.

Examples

A look at two examples to gain conceptual understanding of the strain tensor.

Dilation

Let us look at a small sphere being acted upon by uniform pressure. Let $\mathbf{E} = e\mathbf{1}$ for some nonrotating point. Note that $\mathbf{1}$ is the unit vector. Then $d\mathbf{u} = \mathbf{D}d\mathbf{r}$ reduces to $d\mathbf{u} = e\mathbf{r}$. This result means that the sphere is stretched in all three dimensions by ϵ. Since ϵ is a small number, the fractional volume of the sphere increases by a factor of $3e$. $dV/V = 3e$.

Shearing Strain

Let us look at a shearing strain such that

$$\mathbf{E}= \begin{bmatrix} 0 & \zeta & 0 \\ \zeta & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{6}$$

This gives our elements of the tensor to be $\partial u_1/\partial u_2 = \partial u_2/\partial u_1 = \zeta$. This implies that each axis moves in the direction of the other axis by a factor of ζ:

Refining E

We now know that the diagonal components of $\mathbf{E}$ stretch the small mass along its axes. To find the average stretch, you average the three diagonal elements: $e = (1/3)(\epsilon_1 + \epsilon_2 + \epsilon_3)$

This can be simplified to $e = (1/3)tr\mathbf{E}$, where I have introduced trace (tr), defined as the sum of the diagonal elements of a matrix. Notice that our dilation example above defined e the same way.

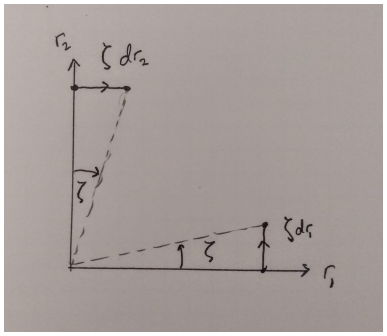

FIG. 2: Shearing Strain

Than $\mathbf{E}$ can be expressed in terms of its spherical part $e\mathbf{1}$, for dilation, and its deviatoric part $\tilde{E}$, measuring deviation from a pure dilation.

$$\mathbf{E} = e1 + \tilde{E} \tag{7}$$

Relating Stress to Strain

We can estimate stress as having a linear relationship with strain. Corresponding with our three surface forces are three relations between stress and strain. Each equation will be of the form $stress = modulus \times strain$.

Tension gives us Young's Modulus.

$$dF/A = YM * dl/l \tag{8}$$

Where A is cross-sectional area, and dl is an extension of length caused by dF.

Pressure gives us the Bulk Modulus.

$$dp = -BM * dV/V \tag{9}$$

This relationships represents that an increase in pressure will decrease volume.

Lastly, shear gives us the Shear Modulus.

$$F/A = SM * dx/dy \tag{10}$$

This relationship represents a displacement of axis relative to each other when under a shearing force.

Hooke's Law

Hooke's Law gives us the relationships between the stress $\mathbf{\Sigma}$ and strain $\mathbf{E}$ tensors. It can be shown that if $\mathbf{\Sigma} = f(\mathbf{E})$, f can be expressed in the general form

$$\mathbf{E} = [3\alpha\mathbf{\Sigma} - (\alpha - \beta)tr\mathbf{\Sigma}1]/(3\ \alpha\beta) \tag{11}$$

Defining Hooke's Law's Constants

Bulk Modulus

Let us think back to our example showing dilation of a sphere. We can evaluate this situation using Hooke's Law. In this case, dilation is caused by pressure p , a uniform stress. We can define the stress tensor as $\Sigma = -p1$, and $tr\mathbf{\Sigma} = -3p$ Substituting these into Hooke's Law and simplifying, we arrive at

$$\mathbf{E} = -(p/\alpha)1$$

Since we are looking at a pure dilation, $\mathbf{E}$ also takes the form

$$\mathbf{E} = e1$$

We look to our example, where we found that

$$dV/V = 3e$$

Combining these three equations, we find $p = -(1/3)\alpha dV/V$, Which takes the form of $dp = -BMdV/V$. We have shown that

$$\alpha = 3BM$$

Shear Modulus

Now we look back at our shearing strain example. Because $\mathbf{E}$ is antisymmetric, it has zero trace, and we can simplify Hooke's Law to

$$\mathbf{E} = \mathbf{\Sigma}/\beta.$$

We again have solvable equations.

$$F/A = \sigma_{12} = \beta\zeta = \beta\epsilon_{12}$$

We use the shear modulus

$$F/A = SMdy/dx = 2SM\zeta$$

Note: We must multiply ζ by two because both axes are tilted by that amount. This leaves us with

$$\beta = 2SM$$

THE PROBLEM:

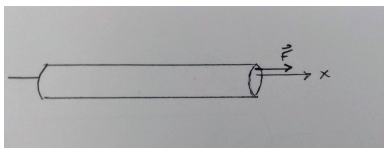

FIG. 3: Thin Metal Wire

"Consider a thin wire which lies along the x-axis. A force is pulling on the end of the wire in the x-direction, stretching the wire."

PART A

"As you have a force purely in the x direction, your stress tensor will have some $_{11} > 0$ and all other $_{ij} = 0$. Convince yourself this is true and then use equation (16.95) from your book to write down the corresponding strain tensor E."

In this case, $\mathbf{F}$ points exclusively along the x axis. The key to this tensor is to recognize that d$\mathbf{A}$ is formed of a scalar area element dA oriented by $\mathbf{n}$.

$$\mathbf{F}(\mathbf{n}_1)dA) = \sigma_{11}dA \text{ Where } \mathbf{\Sigma} = \sigma_{11}$$

We recognize that $tr\mathbf{\Sigma} = \sigma_{11}/3$. Now we can find the strain tensor using Hooke's Law.

$$\mathbf{E} = [3\alpha\boldsymbol{\sigma_1}\mathbf{1} - (\alpha - \beta)\sigma_{11}/3\mathbf{1}]/(3\alpha\beta)$$

Which reduces to

$$\mathbf{E} = \sigma_{11}/(3\alpha\beta)\begin{bmatrix} 2\alpha+\beta & 0 & 0 \\ & \beta-\alpha & 0 \\ 0 & 0 & \beta-\alpha \end{bmatrix}$$

PART B

"From the definition of Young's modulus in equation (16.55), show that $YM = \sigma_{11}/\epsilon_{11}$"

$$stress = modulus \times strain$$
$$dF/A = YM * dl/l$$

It is important to look at the dl/l element. Because it is along the x axis, the strain we use for Young's Modulus is the element $\mathbf{E}_{11}$

$$\sigma_{11} = YM * [\sigma_{11}/(3\alpha\beta)(2\alpha+\beta)]$$
$$\sigma_{11} = YM * \epsilon_{11}$$
$$YM = \sigma_{11}/\epsilon_{11}$$

PART C

"Use both your results to show that the Young's modulus can be written in terms of the Bulk modulus and Shear modulus"

We recycle the equation from above:

$$\sigma_{11} = YM * [\sigma_{11}/(3\alpha\beta)(2\alpha+\beta)]$$

and reduce it to

$$YM = 3\alpha\beta/(2\alpha+\beta)$$

Now we plug in $\alpha = 3BM$ and $\beta = 2SM$ to write Young's modulus as:

$$YM = 9BM * SM/(3BM + SM).$$

This means we only ever need to be given two out of three moduli to solve for the third.

PART D

"As the wire stretches in the x-direction, it actually contracts in the y and z directions. The ratio of the contracting to the stretching is called Poisson's ratio, and denoted with a ."

We know that **E** is the same magnitude in the y and z directions, so ν will be the same ratio for each. Lets look at the surface force $F(d\mathbf{A}) = ma$. Then we decide to shrink the section of mass we are looking at by a factor of λ. Since our surface forces are based on area, they must shrink by a factor of λ^2. However, mass is based on volume, so it shrinks by a cubed factor instead. Dividing out λ^2 on each side, we get

$$F(d\mathbf{A}) = \lambda \mathrm{ma}$$

If we let λ approach zero, we see that our surface forces $F(d\mathbf{A}_1) + F(d\mathbf{A}_2) + F(d\mathbf{A}_3) = 0$. Using the stress tensor we know for the wire,

$$\epsilon_{11} dA = -\epsilon_{22} dA - \epsilon_{33} dA.$$

This relationship shows that a positive change in area on the x axis will result in a negative one on both the y and z axes, and so we can form a ratio with σ, which we call the Poisson Ratio:

$$\nu = -\epsilon_{22}/\epsilon_{11}.$$

PART E

"Express as a function of the Bulk and Shear Moduli".

$$\nu = -\epsilon_{22}/\epsilon_{11} \; \nu = -(\beta - \alpha)/(2\alpha + \beta)$$

Substitute in moduli for α and β

$$\nu = (3BM - 2SM)/(6BM + 2SM)$$

PART F

"Calculate Poisson's ratio for the materials:"

Material	BM	SM	YM	ν
Iron	100	70	91	0.216
Steel	160	84	200	0.277
Copper	140	42	110	0.364
Aluminum	70	24	70	0.346
Lead	7.7	5.6	16	0.207

A wire made out of a softer material like lead will have a smaller Poisson's ratio, because it will be easier to manipulate.